

MACHINE LEARNING

REGRESSION MODELS

NGO QUOC VIET-2025

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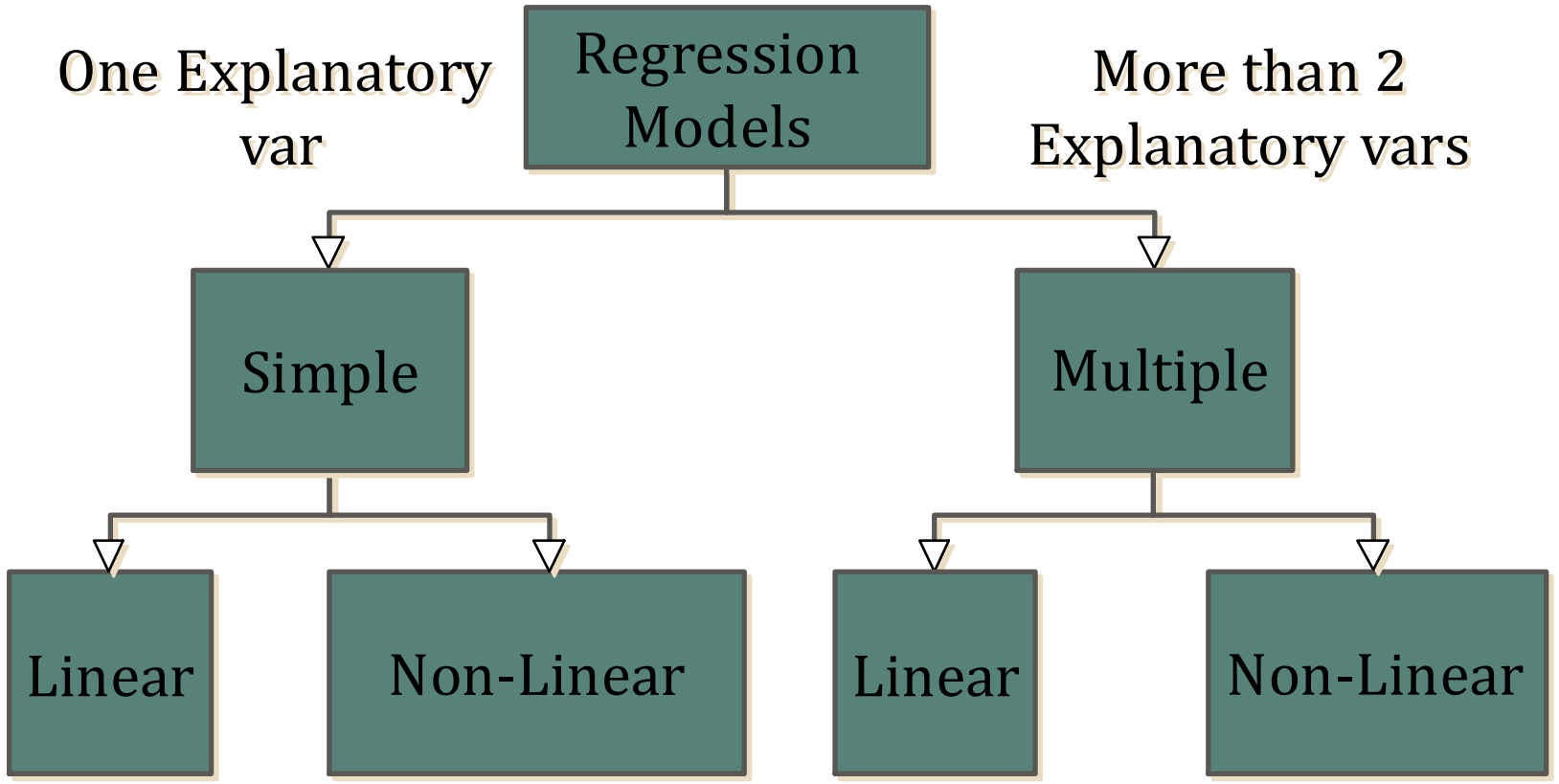
REGRESSION METHOD

- Regression: supervised learning algorithm.
- Represents the relationship between a **dependent variable (or more)** and condition variables (also called explanatory variables).
- Regression models are suitable for prediction and estimation problems.
- Popular regression models: Linear, Logistic, Polynomial, Stepwise, Ridge, Lasso, Elastic Net.

REGRESSION'S USAGE

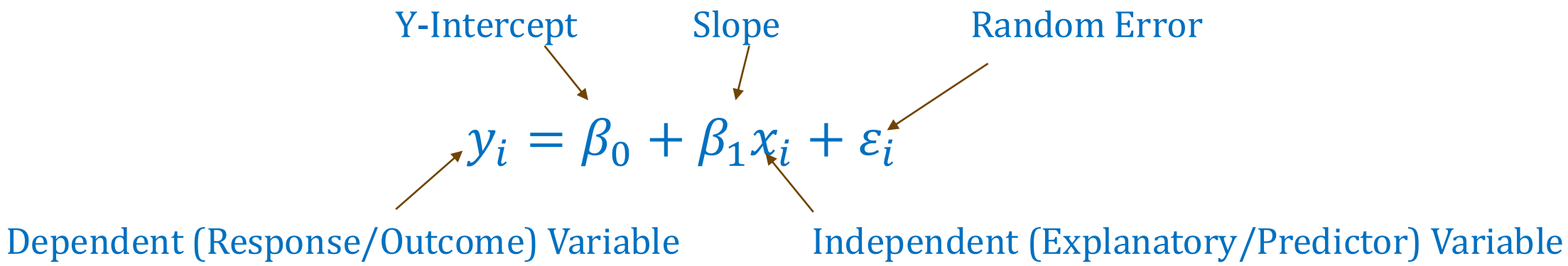
- Forecasting the effects or impact of specific changes. For example, if a manufacturing company wants to forecast how many units of a particular product they need to produce in order to meet the current demand.
- Forecasting trends and future values. For example, how much will the stock price of Lufthansa be in 6 months from now?
- Determining the strength of different predictors—or, in other words, assessing how much of an impact the independent variable(s) has on a dependent variable. For example, if a soft drinks company is sponsoring a football match, they might want to determine if the ads being displayed during the match have accounted for any increase in sales

TYPES OF REGRESSION MODELS



SIMPLE LINEAR REGRESSION

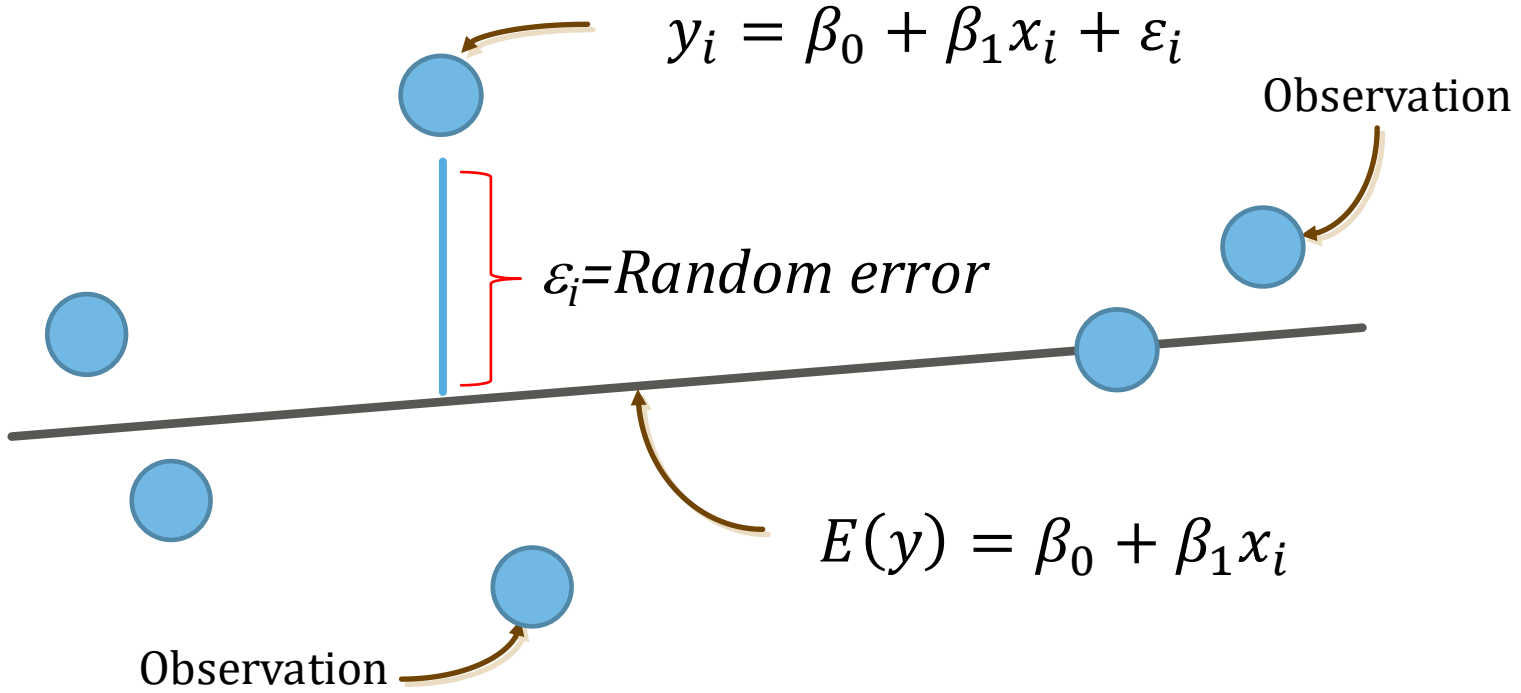
- The SLR model asserts the values of X and Y for individuals in a population are related as follow



- The line $y = \beta_0 + \beta_1 x$ is called the **population regression line.**
- The value of ε, called the error or the noise, varies from observation to observation, follows a normal distribution
- **Goal: Find the values of $\boldsymbol{\beta_0}$ and $\boldsymbol{\beta_1}$ that give the 'best fitting line'.** Best fit: The values of $\boldsymbol{\beta_0}$ **and** $\boldsymbol{\beta_1}$ that minimize the total squared error

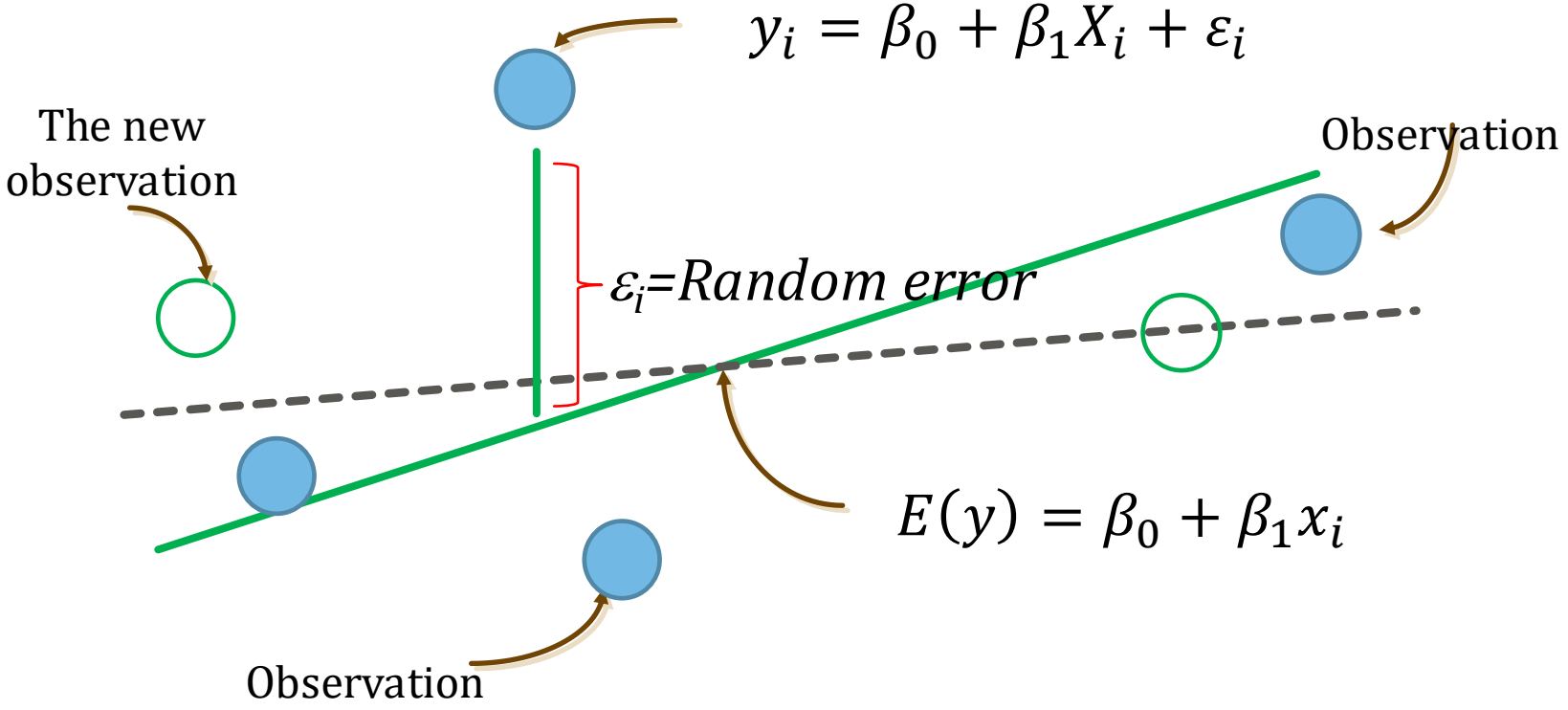
SIMPLE LINEAR REGRESSION

- Given the samples(x_i, y_i), it is necessary to determine the linear function that best represents the relationship between the dependent variable and the condition variables (independent/predictor).



SIMPLE LINEAR REGRESSION

- Determine the linear line with less error



SIMPLE LINEAR REGRESSION

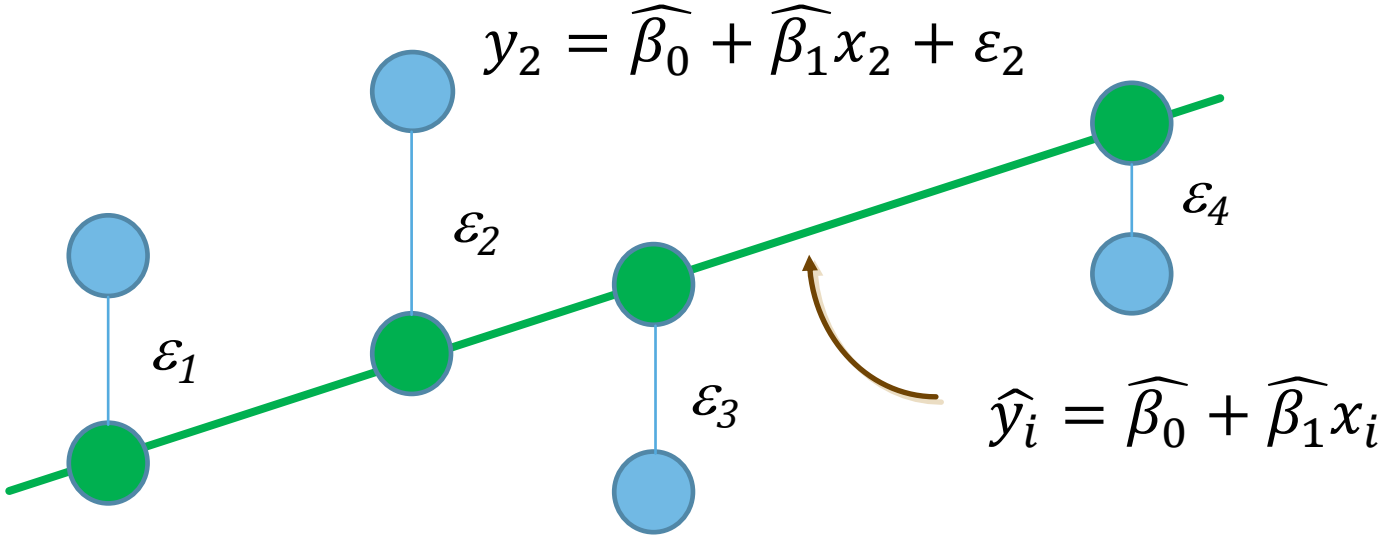
- Determine the linear line that best fits the observed patterns.
- Error of sample point relative to linear line= distance from point to line.
- Best fit: the difference between the real Y value and the Y value lying on the model line is the smallest.
- Using the least squares error.

$$\sum_{i=1}^{n}(y_i - \hat{y}_i)^2 = \sum_{i=1}^{n}\varepsilon_i^2$$

- To look for linear line that best fits the observed patterns.

PARAMETER ESTIMATION

$$LS(\beta_0, \beta_1) = \min \sum_{i=1}^{n} \hat{\varepsilon}_i^2 = \min \sum_{i=1}^{n} \left(y_i - \widehat{\beta_0} - \widehat{\beta_1} x_i\right)^2$$



PARAMETER ESTIMATION

- Take the derivative with respect to β_0, β_1 to determine the minimum

$$\sum_{i=1}^{n} \varepsilon_i^2 = \sum_{i=1}^{n} (y_i - \beta_0 - \beta_1 x_i)^2$$

$$0 = \frac{\partial \sum_{i=1}^{n} \varepsilon_i^2}{\partial \beta_0} = \frac{\partial \sum_{i=1}^{n} (y_i - \beta_0 - \beta_1 x_i)^2}{\partial \beta_0}$$

$$= -2(n\bar{y} - n\beta_0 - n\beta_1 \bar{x})$$

$$\Rightarrow \hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$$

PARAMETER ESTIMATION

$$0 = \frac{\partial \sum_{i=1}^{n} \varepsilon_i^2}{\partial \beta_1} = \frac{\partial \sum_{i=1}^{n} (y_i - \beta_0 - \beta_1 x_i)^2}{\partial \beta_1}$$

$$= -2 \sum_{i=1}^{n} x_i (y_i - \beta_0 - \beta_1 x_i)$$

$$= -2 \sum_{i=1}^{n} x_i (y_i - \bar{y} + \beta_1 \bar{x} - \beta_1 x_i)$$

$$\beta_1 \sum_{i=1}^{n} x_i (x_i - \bar{x}) = \sum_{i=1}^{n} x_i (y_i - \bar{y})$$

$$\Rightarrow \beta_1 = \frac{\sum_{i=1}^{n} (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^{n} (x_i - \bar{x})^2}$$

PARAMETER ESTIMATION

$$\hat{\beta}_1 = \frac{SXY}{SXX}$$

$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$$

$$\bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i \, , \bar{y} = \frac{1}{n}\sum_{i=1}^{n} y_i$$

$$SXY = \sum_{i=1}^{n} (x_i - \bar{x})(y_i - \bar{y})$$

$$SXX = \sum_{i=1}^{n} (x_i - \bar{x})^2$$

EXAMPLE OF LINEAR REGRESSION

- The relationship between the results of the 'competency' test and the learning outcomes of the statistics course.

Student	x_i	y_i	$x_i - \bar{x}$	$y_i - \bar{y}$	$(x_i - \bar{x})^2$	$(y_i - \bar{y})^2$	$(x_i - \bar{x})(y_i - \bar{y})$
	95	85	17	8	289	64	136
	85	95	7	18	49	324	126
	80	70	2	-7	4	49	-14
	70	65	-8	-12	64	144	96
	60	70	-18	-7	324	49	126
Sum	390	385			730	630	470
Mean	78	77					

$$\beta_1 = \frac{\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^{n}(x_i - \bar{x})^2}$$

$$= \frac{470}{730} = 0.644$$

$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1\bar{x} = 77 - \frac{0.644}{78}$$

$$= 26.768$$

$$\Rightarrow \hat{y} = 26.768 + 0.644 * x$$

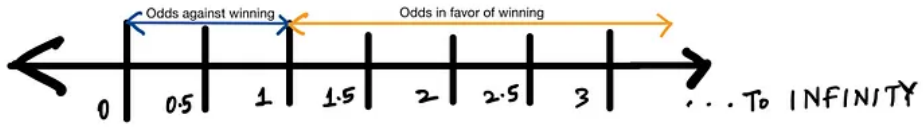
If the student takes the test ability test and gets 80 points, the final score may be 78.288

SOME BASIC PROBABILITY CONCEPTS

- Given p be a probability of an event. **Odds** are the ratio of something happening to something not happening

$$\boldsymbol{odds = p/(1-p)}$$

  - Example of odds between wining and losing: $win = 4, loses = 6 \rightarrow odds = 0.66$; $wins = 4, los$ 0.0081; $wins = 94, loses = 6 \rightarrow odds = 15.666$; $wins = 494, loses = 6 \rightarrow odds = 82.33$; *The odds in favor of winning range between 1 and infinity.*



- Taking the log of odds make it look symmetrical. Log(odds) helps getting a Normal distribution.

$$\boldsymbol{logit = log(odds)}: \boldsymbol{logit(p) = log\big(p/(1-p)\big)}$$

- **log odds** are an alternate way of expressing probabilities.
- The ratio of two odds: $\boldsymbol{OR = \frac{odds_1}{odds_2}}$

SOME BASIC PROBABILITY CONCEPTS

- Odds ratio (OR): ratio of two odds

$$OR = \frac{odds_1}{odds_2}$$

- Example:

	Cancer	Non cancer
Smoking	647	622
No smoking	2	27

Odds (smoking in cancer group)= 647/2 = 323.5

Odds (smoking in the non-cancer) = 622/27= 23.03

Odds ratio = 323.5/23.03 = 14.1 (The odds of smoking in cancer group is 14 times the odds of smoking in the non-cancer)

WHAT IS LOGISTIC REGRESSION?

- Logistic regression is a classification algorithm used to find the probability of event success and event failure
- It is mostly used when the dependent variable is binary (0/1, True/False, Yes/No). Meanwhile, the independent variables can be *continuous* (eg: temperature) or discrete (ordinal or nomial).
- It supports categorizing data into discrete classes by studying the relationship from a given set of labelled data
- It learns a linear relationship from the given dataset and then introduces a non-linearity in the form of the **Sigmoid** function

https://www.geeksforgeeks.org/advantages-and-disadvantages-of-logistic-regression/

LOGISTIC REGRESSION ASSUMPTIONS

- The dependent variable is binary. *But there are some other types of logistic regression*
- There should be no, or very little, multicollinearity between the predictor variables.
- The independent variables should be linearly related to the log odds
- Logistic regression requires fairly large sample sizes

LOGISTIC REGRESSION BASED ON LOGIT

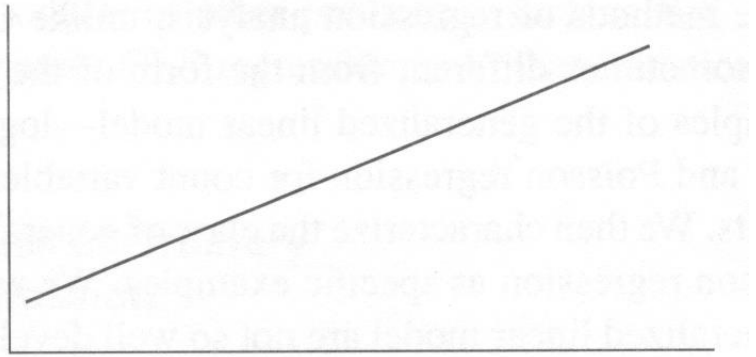
- The independent variables should be linearly related to the log odds

$$logit(p) = ln\left(\frac{p}{1-p}\right) = \beta_0 + \beta X$$

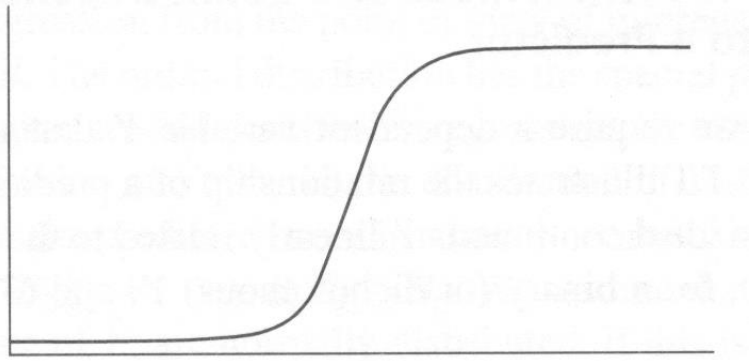
- The estimated probability is

$$p = \frac{e^{\beta_0+\beta X}}{1 + e^{-(\beta_0+\beta X)}}$$

- If $\beta_0 + \beta X = 0$, then $p = 0.5$.
- As $\beta_0 + \beta X$ gets really big, then p approaches 1
- As $\beta_0 + \beta X$ gets really small, then p approaches 0
- The model can also be described using linear algebra, with a vector for the coefficients (Beta) and a matrix for the input data (X) and a vector for the output (y), $y = X\beta$.



$$logit(p) = \beta_0 + \beta X$$



$$p = \frac{e^{\beta_0+\beta X}}{1 + e^{-(\beta_0+\beta X)}}$$

LOGISTIC REGRESSION

- Denote

$$S(x_i\beta) = P(Y_i = 1|x_i) = \frac{1}{1 + e^{-\beta x}}$$

- The output is interpreted as a probability from a Binomial probability distribution function for the class labeled 1, if the two classes in the problem are labeled 0 and 1.
- In logistic function, x_i is a $1xK$ vector of inputs and β is a $Kx1$ vector of coefficients
- In logit form

$$logit\big(S(x_i\beta)\big) = log\left(\frac{S(x_i\beta)}{1 - S(x_i\beta)}\right)$$
$$= \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \cdots$$

PARAMETER ESTIMATION OF LOGISTIC REGRESSION

- Two common frameworks to estimate the parameters are
  - Maximum Likelihood Estimation.
  - Least Squares Optimization (iteratively reweighted least squares).
- And other frameworks: Newton-Raphson method, Covariance matrix of the estimator
- Likelihood is determined by

$$L(\beta; y, x) = \prod_{i=1}^{n} S(x_i\beta)^{y_i}\big(1 - S(x_i\beta)\big)^{1-y_i}$$

$$= \prod_{i=1}^{n} \frac{y_i e^{\beta_0+\beta_i x_i}}{1 + e^{\beta_0+\beta_i x_i}}$$

https://www.statlect.com/fundamentals-of-statistics/classification-models

PARAMETER ESTIMATION OF LOGISTIC REGRESSION

- Denote the $Nx1$ vector of all outputs by y and the NxK matrix of all inputs by X.
- The **log-likelihood** of the logistic model is

$$l(\beta; y, x) = \sum_{i=1}^{N} [-ln(1 + exp(\beta_i x_i) + y_i x_i \beta)]$$

- The **score vector**: the vector of first derivatives of the log-likelihood with respect to the parameter β.

$$\nabla_\beta l(\beta; y, x) = \sum_{i=1}^{N} [y_i - S(x_i \beta)] x_i$$

- The **Hessian**: the matrix of second derivatives

$$\nabla_{\beta\beta} l(\beta; y, x) = -\sum_{i=1}^{N} x_i^T x_i S(x_i \beta)[1 - S(x_i \beta)]$$

https://realpython.com/logistic-regression-python/

PARAMETER ESTIMATION OF LOGISTIC REGRESSION

- The maximum likelihood estimator $\widehat{\theta}$ of the parameter θ solves

$$\hat{\beta} = arg \max_{\beta} l(\beta; y, x)$$

- In general, there is no analytical solution of this maximization problem and a solution must be found numerically (https://www.statlect.com/fundamentals-of-statistics/maximum-likelihood-algorithm).
- The maximization problem has a solution, and at the maximum the score vector satisfies the first order condition

$$\nabla l(\beta; y, x) = \sum_{i=1}^{N} [y_i - S(x_i\beta)]x_i = 0$$

- The first order condition above has no explicit solution. In most statistical software packages it is solved by using the Newton-Raphson method (https://web.mit.edu/10.001/Web/Course_Notes/NLAE/node6.html).

PARAMETER ESTIMATION USING NEWTON-RAPHSON METHOD

- The Newton-Raphson method is pretty simple: starting from an initial solution, e.g. $\hat{\beta}_0 = 0$, and recursively updating the solution with the equation

$$\hat{\beta}_t = \hat{\beta}_{t-1} - \left[\nabla_{\beta\beta} l\left(\hat{\beta}_{t-1}; y, x\right)\right]^{-1} \nabla_{\beta} l\left(\hat{\beta}_{t-1}; y, x\right)$$

  until numerical convergence

- Denote $\hat{y}_t$, the $Nx1$ vector of conditional probabilities of the outputs computed by using $\hat{\beta}_t$.

$$\hat{y}_t = \left[S\left(x_1 \hat{\beta}_t\right) \cdots S\left(x_N \hat{\beta}_t\right)\right]^T$$

PARAMETER ESTIMATION USING NEWTON-RAPHSON METHOD

- Denote W_t, the NxN diagonal matrix

$$\begin{bmatrix} S(x_1\hat{\beta}_t)\left(1-S(x_1\hat{\beta}_t)\right) & 0 & \dots & 0 \\ 0 & S(x_2\hat{\beta}_t)\left(1-S(x_2\hat{\beta}_t)\right) & \cdots & 0 \\ \vdots & \vdots & \cdots & S(x_N\hat{\beta}_t)\left(1-S(x_N\hat{\beta}_t)\right) \\ 0 & 0 & & \end{bmatrix}$$

- The NxK diagonal matrix of inputs

$$X = [x_1 \cdots x_N]^T$$

- The score in Newton-Raphson recursive formula can be written as

$$\nabla_\beta l(\hat{\beta}_{t-1}; y, x) = X^T(y - \hat{y}_{t-1})$$

$$\nabla_{\beta\beta} l(\beta; y, x) = -X^T W_{t-1} X$$

- Therefore, the Newton-Raphson formula becomes

$$\hat{\beta}_t = \hat{\beta}_{t-1} + (X^T W_{t-1} X)^{-1} X^T (y - \hat{y}_{t-1})$$

PARAMETER ESTIMATION USING ITERATIVELY REWEIGHTED LEAST SQUARES

- The Newton-Raphson formula above is equivalent to the Iteratively Reweighted Least Squares (IRLS) formula

$$\hat{\beta}_t = (X^T W_{t-1} X)^{-1} X^T W_{t-1} \left[X\hat{\beta}_{t-1} + W_{t-1}^{-1}(y - \hat{y}_{t-1}) \right]$$

- The above formula is obtained by performing a Weighted Least Squares (WLS) estimation with weights W_{t-1} of a linear regression of the dependent variables $X\hat{\beta}_{t-1} + W_{t-1}^{-1}(y - \hat{y}_{t-1})$ on the regressors X.

POPULAR LOSS FUNCTIONS IN REGRESSION

Prediction output: $\hat{y}_i = f_w(x_i)$

Mean Square Error/Quadratic Loss/L2 Loss:

$$MSE = \sum_{i=1}^{n} \frac{(y_i - \hat{y}_i)^2}{n}$$

Mean Absolute Error/L1 Loss

$$MAE = \sum_{i=1}^{n} \frac{|y_i - \hat{y}_i|}{n}$$

Mean Bias Error/L1 Loss

$$MBE = \sum_{i=1}^{n} \frac{(y_i - \hat{y}_i)}{n}$$

EVALUATION OF REGRESSION MODEL

- Use RMSE (Root Mean Square Error) to determine the error between predicted values and actual observed values

$$RMSE = \sqrt{\sum_{i=1}^{n} \frac{(\hat{y}_i - y_i)^2}{n}}$$

LOGISTIC REGRESSION – PYTHON CODE LINKS

- https://scikit-learn.org/stable/modules/generated/sklearn.linear_model.LogisticRegression.html
- https://realpython.com/logistic-regression-python/

SUMMARY

- A regression is a statistical technique that relates a dependent variable to one or more independent (explanatory) variables
- A regression model is able to show whether changes observed in the dependent variable are associated with changes in one or more of the explanatory variables
- Linear and logistic regressions are easy to understand and install.

THANK YOU